

Spring Data JPA - 3 Week Detailed Roadmap

Week 1: Spring Data JPA Fundamentals

1. Introduction to JPA & ORM

- What is ORM?
- Why ORM is needed?
- JDBC vs JPA
- Hibernate vs JPA
- Benefits of JPA
- JPA Architecture
- Persistence Context
- Entity Lifecycle States (Transient, Persistent, Detached, Removed)

2. Setting Up Spring Data JPA

- Spring Boot Starter Data JPA
- Database Configuration
- MySQL Configuration
- H2 Database
- DataSource Configuration
- application.properties Setup

3. Entity Mapping

- @Entity
- @Table
- @Id
- @GeneratedValue
- Generation Types
- @Column
- @Transient
- @Lob

4. Repository Layer

- CrudRepository
- PagingAndSortingRepository
- JpaRepository
- save()
- findById()

- findAll()
- deleteById()
- deleteAll()

5. CRUD Application

- Create Operation
- Read Operation
- Update Operation
- Delete Operation
- REST API Integration
- Testing CRUD APIs

Week 2: Relationships & Queries

6. Relationships in JPA

- One-to-One Mapping
- @OneToOne
- @JoinColumn
- One-to-Many Mapping
- @OneToMany
- Many-to-One Mapping
- @ManyToOne
- Many-to-Many Mapping
- @ManyToMany
- @JoinTable
- Cascade Types
- Fetch Types (EAGER & LAZY)

7. Derived Query Methods

- findByName()
- findByEmail()
- findByAgeGreaterThan()
- findBySalaryBetween()
- findByNameContaining()
- findByNameStartingWith()
- findByNameEndingWith()
- findOrderBySalaryDesc()

8. JPQL

- Introduction to JPQL
- @Query Annotation
- Named Parameters
- Positional Parameters
- SELECT Query
- UPDATE Query
- DELETE Query

9. Native SQL Queries

- Native Query Basics
- @Query(nativeQuery=true)

- Insert Query
- Update Query
- Delete Query
- Custom SQL Queries

10. Sorting & Pagination

- Sort Object
- Page Object
- Pageable Interface
- PageRequest
- Multi-column Sorting
- Pagination in REST APIs

Week 3: Advanced Spring Data JPA

11. Entity Relationships Deep Dive

- Bidirectional Mapping
- Unidirectional Mapping
- mappedBy Attribute
- Join Tables
- Owning Side vs Inverse Side

12. Auditing

- @CreateDate
- @LastModifiedDate
- @CreatedBy
- @LastModifiedBy
- EnableJpaAuditing

13. Transaction Management

- ACID Properties
- @Transactional
- Propagation Types
- Isolation Levels
- Rollback Handling

14. Specifications API

- Specification Interface
- CriteriaBuilder
- CriteriaQuery
- Predicate
- Dynamic Queries

15. Projections

- Interface-based Projection
- DTO Projection
- Constructor Projection

16. EntityManager

- persist()
- merge()

- remove()
- flush()
- detach()
- clear()

17. Caching

- First-Level Cache
- Second-Level Cache
- Hibernate Cache
- Cache Configuration

18. Locking

- @Version
- Optimistic Locking
- Pessimistic Locking
- Lock Modes

19. Interview Topics

- JPA vs Hibernate
- save() vs saveAndFlush()
- getByName() vs findById()
- LAZY vs EAGER
- Cascade Types
- orphanRemoval
- N+1 Query Problem
- Persistence Context
- Dirty Checking
- First-Level Cache
- @Transactional Internals

Final Mini Project

Build an Employee Management System or Student Management System using Spring Boot, Spring Data JPA, MySQL, REST APIs, Validation, Pagination, Sorting, Relationships, JPQL Queries, Auditing, and Transaction Management.